

Physiology

CEREBRAL CIRCULATION AND CSF FLUID

CEREBRAL CIRCULATION

It is the movement of blood through a network of cerebral arteries and veins supplying the brain. The rate of cerebral blood flow in an adult human is typically 750 milliliters per minute, or about 15% of cardiac output. The volume of blood in circulation is called the cerebral blood flow. This equates to an average perfusion of 50 to 54 millilitres of blood per 100 grams of brain tissue per minute. CBF is tightly regulated to meet the brain's metabolic demands.

Too much blood (a condition known as hyperemia) can raise intracranial pressure (ICP), which can compress and damage delicate brain tissue. Too little blood flow (ischemia) results if blood flow to the brain is below 18 to 20 ml per 100 g per minute, and tissue death occurs if flow dips below 8 to 10 ml per 100 g per minute. In brain tissue, a biochemical cascade known as the ischemic cascade is triggered when the tissue becomes ischemic, potentially resulting in damage to and the death of brain cells.

Factors Determining CBF

Cerebral blood flow is determined by a number of factors, such as viscosity of blood, how dilated blood vessels are, and the net pressure of the flow of blood into the brain, known as cerebral perfusion pressure which is determined by the body's blood pressure. Cerebral perfusion pressure (CPP) is defined as the mean arterial pressure (MAP) minus the intracranial pressure (ICP). In normal individuals, it should be above 50

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mm Hg. Intracranial pressure should not be above 15 mm Hg (ICP of 20 mm Hg is considered as Intracranial Hypertension.) Cerebral blood vessels are able to change the flow of blood through them by altering their diameters in a process called autoregulation; they constrict when systemic blood pressure is raised and dilate when it is lowered. Arterioles also constrict and dilate in response to different chemical

concentrations. For example, they dilate in response to higher levels of carbon dioxide in the blood and constrict in response to lower levels of carbon dioxide.

For example, assuming a person with an arterial partial pressure of carbon dioxide (PaCO₂) of 40 mmHg (normal range of 38–42 mmHg) and a CBF of 50 ml per 100g per min. If the PaCO₂ dips to 30 mmHg, this represents a 10 mmHg decrease from the initial value of PaCO₂. Consequently, the CBF decreases by 1ml per 100g per min for each 1mmHg decrease in PaCO₂, resulting in a new CBF of 40ml per 100g of brain tissue per minute. In fact, for each 1 mmHg increase or decrease in PaCO₂, between the range of 20–60 mmHg, there is a corresponding CBF change in the same direction of approximately 1–2 ml/100g/min, or 2–5% of the CBF value. This is why small alterations in respiration pattern can cause significant changes in global CBF, specially through PaCO₂ variations.

CBF is equal to the cerebral perfusion pressure (CPP) divided by the cerebrovascular resistance (CVR):

$$\text{CBF} = \text{CPP} / \text{CVR}$$

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Control of CBF is considered in terms of the factors affecting CPP and the factors affecting CVR. CVR is controlled by four major mechanisms:

1. Metabolic control (or 'metabolic autoregulation')
2. Pressure autoregulation
3. Chemical control (by arterial pCO₂ and pO₂)
4. Neural control

Role of intracranial pressure

Increased intracranial pressure (ICP) causes decreased blood perfusion of brain cells by mainly two mechanisms:

1. Increased ICP constitutes an increased interstitial hydrostatic pressure that, in turn, causes a decreased driving force for capillary filtration from intracerebral blood vessels.
2. Increased ICP compresses cerebral arteries, causing increased cerebrovascular resistance (CVR).
3. Increased ICP → ↓ cerebral blood flow.
4. The ICP follows the mean cerebral venous pressure.
5. When the ICP rises to 30 mm Hg, the cerebral blood vessels are markedly compressed and the cerebral blood flow is decreased leading to cerebral ischemia.
6. Cerebral ischemia → Cushing's reflex.

Cerebral perfusion pressure

Cerebral perfusion pressure, or CPP, is the net pressure gradient causing cerebral blood flow to the brain (brain perfusion). It must be maintained within narrow limits

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because too little pressure could cause brain tissue to become ischemic (having inadequate blood flow), and too much could raise intracranial pressure (ICP).

Imaging

Arterial spin labelling and positron emission tomography are neuroimaging techniques that can be used to measure CBF. These techniques are also used to measure regional CBF (rCBF) within a specific brain region. rCBF at one location can be measured over time by thermal diffusion

The blood brain barrier (BBB)

- The low permeability of cerebral capillaries limits the passage of many substances and drugs from the blood to the brain.
- This is called the blood brain barrier (BBB).

CSF

- Cerebrospinal fluid (CSF) is a clear, plasma-like fluid (an ultrafiltrate of plasma) that bathes the central nervous system (CNS). It occupies the central spinal canal, the ventricular system, and the subarachnoid space.
- CSF performs vital functions including: Support; Shock absorber; Homeostasis; Nutrition; Immune function.
- Adult CSF volume is estimated to be 150 ml with a distribution of 125 ml within the subarachnoid spaces and 25 ml within the ventricles.
- CSF is predominantly secreted by the choroid plexus with other sources playing a more poorly defined role, an adult produces between 400 to 600 ml per day.
- The constant secretion of CSF contributes to complete CSF renewal four to five times per 24-hour period in the average young adult.

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- Reduction of CSF turnover may contribute to the accumulation of metabolites seen in aging and neurodegenerative diseases. The composition of CSF is closely regulated, and any variation can be used for diagnostic purposes.
- Out-pouching of the arachnoid mater (arachnoid granulations) are responsible for the resorption of CSF into the dural venous sinuses.
- Disequilibrium in synthesis and resorption or obstruction of circulation results in CSF accumulation and raised intracranial pressure called hydrocephalus.
- CSF be examined clinically through a lumbar puncture. With a lumbar puncture, physicians can look for abnormalities in the CSF, which can be helpful when creating a differential diagnosis.

Structure and Function

1. Support - The CSF supports the weight of the brain estimated at 1500 gm and suspends it in neutral buoyancy to a net weight of about 25 gm. Hence, the entire brain density is cushioned, protecting it from crushing into the bony cranium.
2. Shock absorber - It protects the brain
3. Homeostasis - The biochemical constituents and volume of the CSF play vital cerebral homeostatic roles:
 1. Maintains stable intrinsic CNS temperature
 2. Biochemical constituents and electrolytes maintain the osmotic pressure responsible for normal CSF pressure which is essential to maintaining normal cerebral perfusion
 3. Biochemical waste products diffuse into the CSF and are removed as

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CSF is resorbed through arachnoid granulations into venous circulation, a small percentage of CSF also drains into lymphatic circulation.

4. Nutrition - The CSF contains glucose, proteins, lipids, and electrolytes, providing essential CNS nutrition.
5. Immune function - The CSF contains immunoglobulins and mononuclear cells.

Ventricles and CSF (Figure 1)

- CSF is produced mainly by a structure called the choroid plexus in the lateral, third and fourth ventricles.
- CSF flows from the lateral ventricle to the third ventricle through the interventricular foramen (also called the foramen of Monro).
- The third ventricle and fourth ventricle are connected to each other by the cerebral aqueduct (also called the Aqueduct of Sylvius).
- CSF then flows into the subarachnoid space through the foramina of Luschka (there are two of these) and the foramen of Magendie (only one of these).
- Absorption of the CSF into the blood stream takes place in the superior sagittal sinus through structures called arachnoid villi.
- When the CSF pressure is greater than the venous pressure, CSF will flow into the blood stream. However, the arachnoid villi act as "one way valves"...if the CSF pressure is less than the venous pressure, the arachnoid villi will NOT let blood pass into the ventricular system.

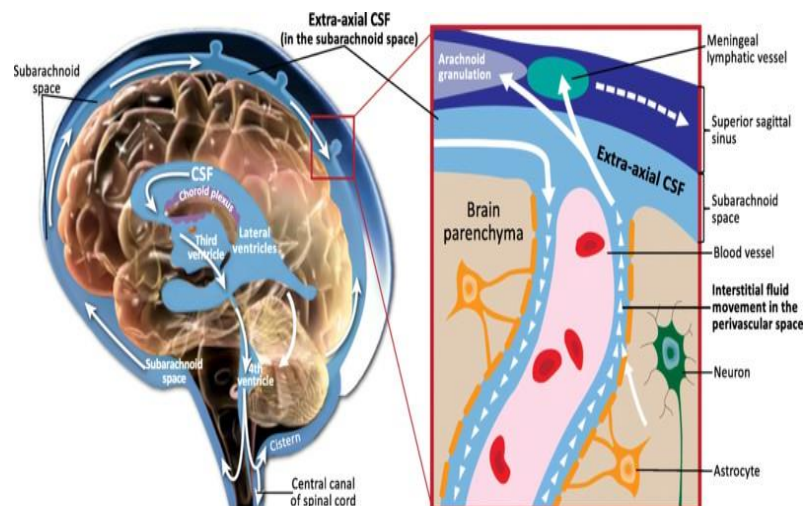


Figure 1: Ventricles and CSF

Blood Supply and Lymphatics

- The choroid plexus is a group of fenestrated blood capillaries located in the ventricular system. The choroid plexus mostly synthesizes CSF.
- Arachnoid granulations are responsible for CSF resorption; they drain CSF into the dural venous sinuses.
- CSF drains into the lymphatic circulation, via lymph ducts contiguous to the olfactory duct, as it passes through the cribriform plate.

Clinical Significance

- Hydrocephalus is a pathological condition of abnormal accumulation of CSF caused by increased CSF production, blockage of flow, or decreased absorption. The ventricles distend in order to accommodate elevated CSF volumes, potentially causing damage to the brain by pressing its tissue against the bony skull. Hydrocephalus may be congenital or acquired.
- CSF Leak is a condition in which CSF is able escape from the subarachnoid space through a hole in the surrounding dura. The volume

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of CSF lost in a leak is very variable, ranging from insignificant to very substantial amounts.

- Syringomyelia due to blockage of CSF circulation.
- Meningitis is a condition in which the coverings of the brain become inflamed. There are two classifications of meningitis: aseptic and bacterial. Aseptic meningitis can be caused by agents such as fungi, medications, and cancer metastasis, but the majority of aseptic meningitis cases are caused by viruses. Fever, nuchal rigidity, and photophobia are classic presenting symptoms. Diagnosis is made via an analysis of CSF obtained through LP.
- Subarachnoid Hemorrhage (SAH) is the leakage of blood into the subarachnoid space where it mixes with the CSF. SAH is most commonly caused by trauma with 80% of nontraumatic SAHs being caused by aneurysm rupture. Other nontraumatic causes of SAH include arteriovenous malformations and vasculitis.

Lumbar Puncture and CSF Analysis: -

- Lumbar puncture is a sterile procedure, done to obtain CSF samples for diagnostic purposes. It involves passing a needle into the subarachnoid space at the levels between L2 and L5 vertebrae. However, most commonly lumbar puncture is performed between L4 and L5. Biochemical, microbiologic and cytologic studies are then carried out on the sample.

Conclusion

Cerebrospinal fluid (CSF) plays an essential role in maintaining the homeostasis of the central nervous system.

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The functions of CSF include: (1) buoyancy of the brain, spinal cord, and nerves; (2) volume adjustment in the cranial cavity; (3) nutrient transport; (4) protein or peptide transport; (5) brain volume regulation through osmoregulation; (6) buffering effect against external forces; (7) signal transduction; (8) drug transport; (9) immune system control; (10) elimination of metabolites and unnecessary substances; and finally cooling of heat generated by neural activity.

For CSF to fulfil these functions, fluid-like movement in the ventricles and subarachnoid space is necessary.

Abnormalities in the CSF flow or introduction of infection and or irritants can seriously affect an individuals functioning.

2. Basic functions of the synapses and neurotransmitters

(I)

The central nervous system contains more than 100 billion neurons (figure 2). Incoming signals enter the neuron through synapses located mostly on the neuronal dendrites, but also on the cell body. For different types of neurons, there may be only a few hundred or as many as 200,000 such synaptic connections from input fibers. Conversely, the output signal travels by way of a single axon leaving the neuron. Then, this axon has many separate branches to other parts of the nervous system or peripheral body.